

Quiz 1 (40 points)
Due: 16/10, Wed., in class.

Let $(\Omega, \mathcal{F}, \mu)$ is a measurable space. Questions are testing: convergence theorems, two fundamental convergence modes.

1. State Fatou's lemma and dominate convergence theorem.
2. Use Fatou's lemma to prove dominate convergence theorem.
3. Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of integrable functions converges a.s. to an integrable function f . Show that

$$\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0 \iff \lim_{n \rightarrow \infty} \int |f_n| d\mu = \int |f| d\mu.$$

Hint: Using the following DCT result in Homework and Fatou's lemma:

Assume that $\{f_n\}_{n=1}^{\infty}, \{g_n\}_{n=1}^{\infty}, f, g$ are integrable and g_n are non-negative, $f_n \rightarrow f$ pointwise, $g_n \rightarrow g$ pointwise, $|f_n| \leq g_n$ a.s., and $\int g_n d\mu \rightarrow \int g d\mu$. Then

$$\int f_n d\mu \rightarrow \int f d\mu.$$

Proof. $(\implies)$: Note that $|f_n(x)| \leq |f(x)| + |f_n(x) - f(x)|$ for all $x \in X$ and all $n \in \mathbb{N}$. Thus

$$\limsup_{n \rightarrow \infty} \int |f_n| d\mu \leq \int |f| d\mu + \limsup_{n \rightarrow \infty} \int |f_n - f| d\mu = \int |f| d\mu$$

Combining this with the inequality from Fatou's Lemma yields the desired convergence.

$(\impliedby)$: Define $g_n := |f_n| + |f|$ and $g := 2|f|$. Then by assumption $\int g_n \rightarrow \int g$. Since $|f_n - f|$ converges to zero μ -almost everywhere and is dominated by g_n , Homework gives

$$\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = \int 0 d\mu = 0$$

The proof is completed. □

4. We defined convergence in probability (measure): $\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| > \varepsilon) = 0$, in general, we have convergence in measure accordingly: $\lim_{n \rightarrow \infty} \mu(|X_n - X| > \varepsilon) = 0$.

We defined convergence almost surely (a.s.): $\mathbb{P}(\lim_{n \rightarrow \infty} X_n = X) = 1$, in general, we have convergence almost everywhere (a.e.): $\mu(\lim_{n \rightarrow \infty} X_n = X) = 1$.

If $p > 0$ and

$$\int |f - f_k|^p d\mu \rightarrow 0$$

as $k \rightarrow \infty$, there is a constant $M > 0$ such that

$$\int |f_k|^p \leq M, \quad \text{for all } k,$$

show that

$$\int |f|^p \leq M.$$

Hint: Start from showing that $f_k \rightarrow f$ in measure.

Proof. Suppose that there is $\varepsilon > 0$ such that $|X| \geq \varepsilon$ for all k , where $E = \{|f - f_k| > \varepsilon\}$. Then

$$\int |f - f_k|^p \geq \int_E |f - f_k|^p \geq \int_E \varepsilon^p \geq \varepsilon^{p+1} > 0$$

for all ε . Hence, we have $f_k \rightarrow f$ in measure.

Next, we use the previous problem, we know that $f_k \rightarrow f$ in measure; in particular, there exists a subsequence $f_{k_j} \rightarrow f$ a.e.. Thus, by Fatou's lemma,

$$\int |f|^p = \int \liminf |f_{k_j}|^p \leq \liminf \int |f_{k_j}|^p \leq M,$$

the proof is completed. □